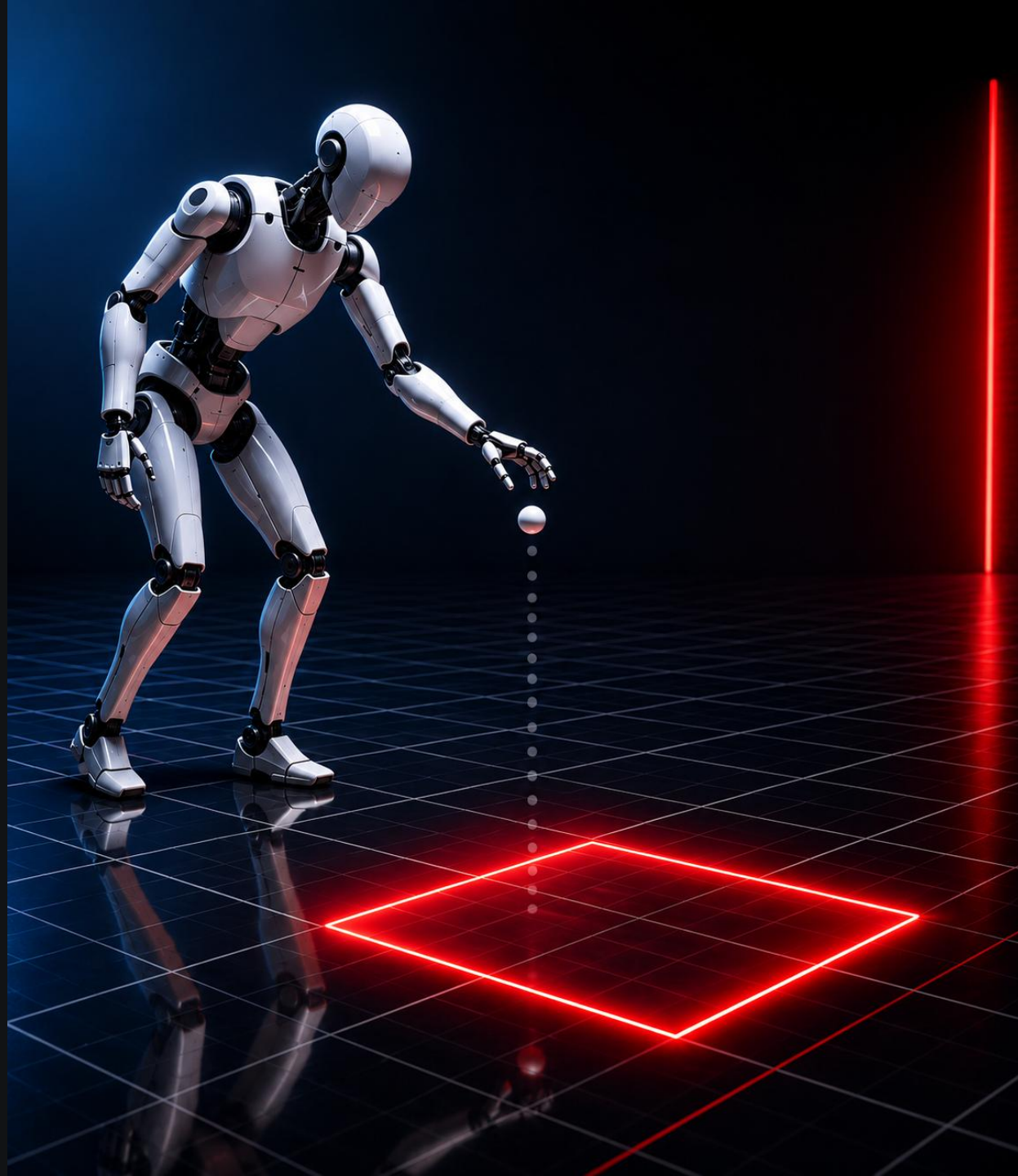


"I think it should be easy."

From Baseline Control to Reinforcement Learning

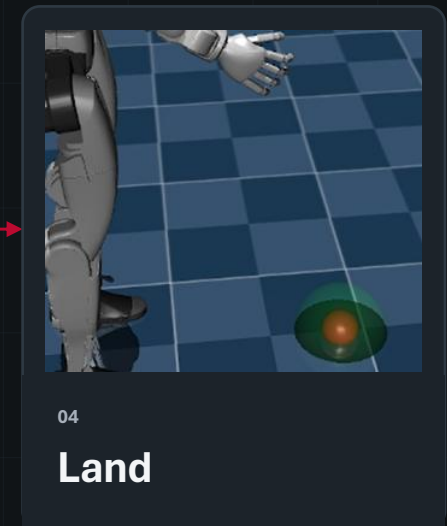
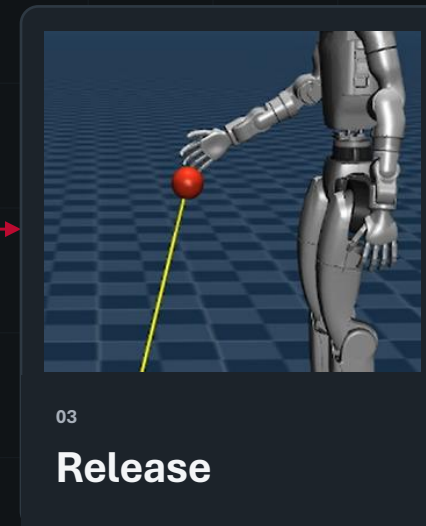
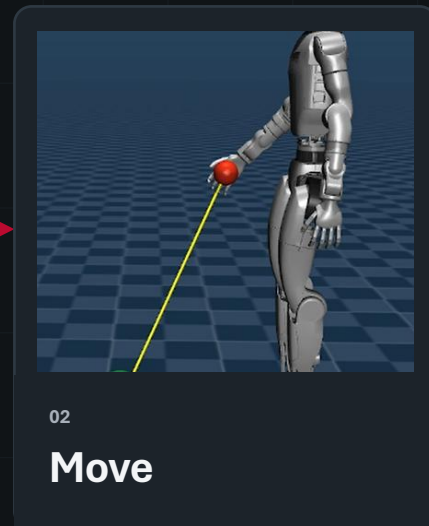
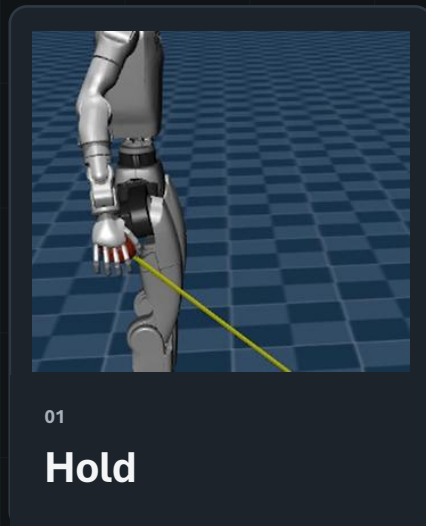
Audi Development Camp 2026_Team 03

Andrea
Francesco
Uulkan
Aiman
Tianyu
—Letian



How Difficult Can One Simple Task Be?

The visible task is small. The hidden control problem is not.



A small timing error becomes a landing error.

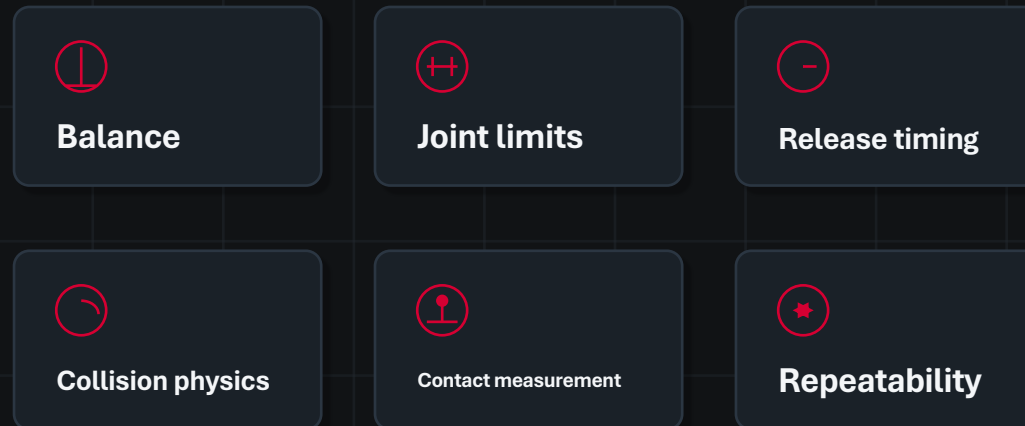
What Could Go Wrong?

Timing error → landing error

29 DoF

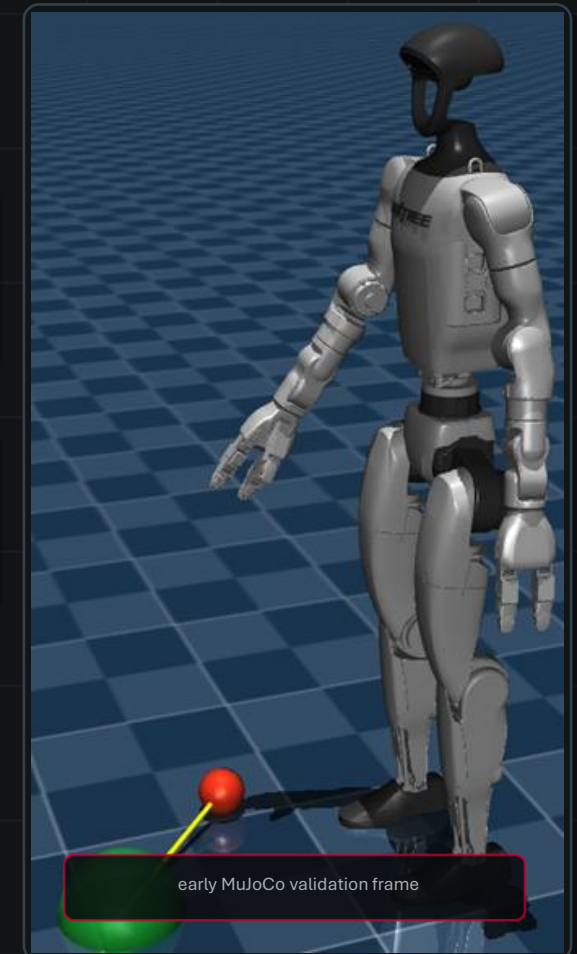
humanoid control problem

One visible task hides a full-body control problem.



timing error

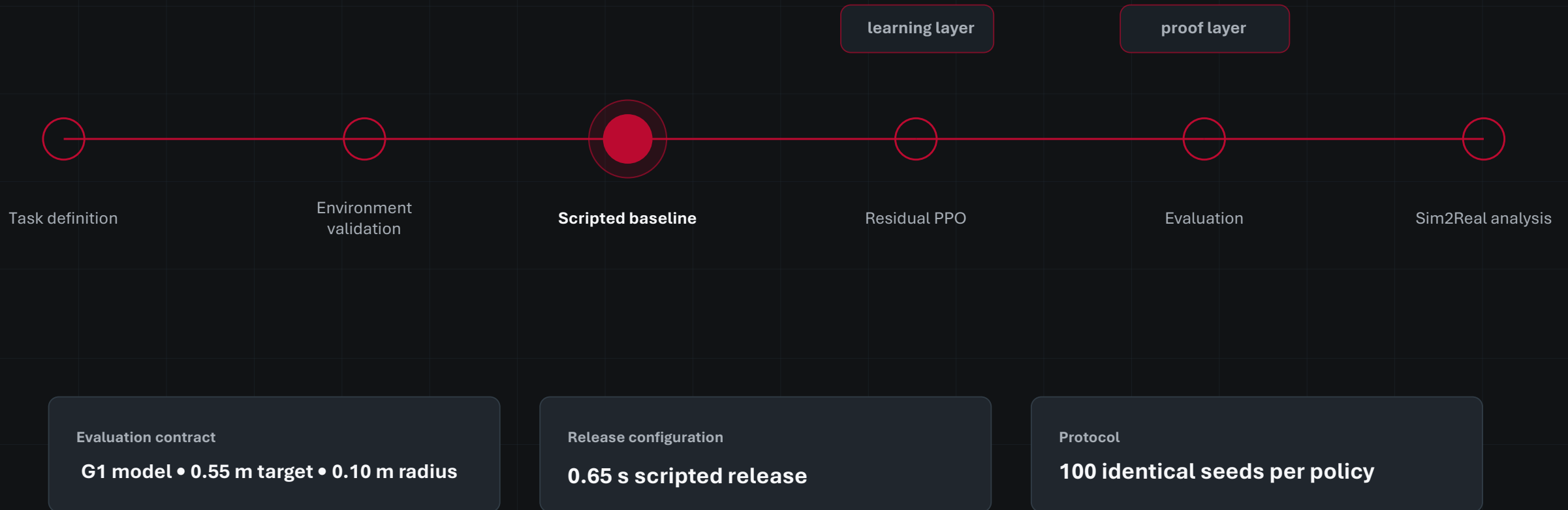
landing error



Can we trust the environment enough to train anything at all?

We Did Not Start With AI

Trust before learning.



LVL02: Building a Simulator We Could Trust

Validated environment contract.

Unitree G1

29 DoF

Ball

radius 4 cm

mass 50 g

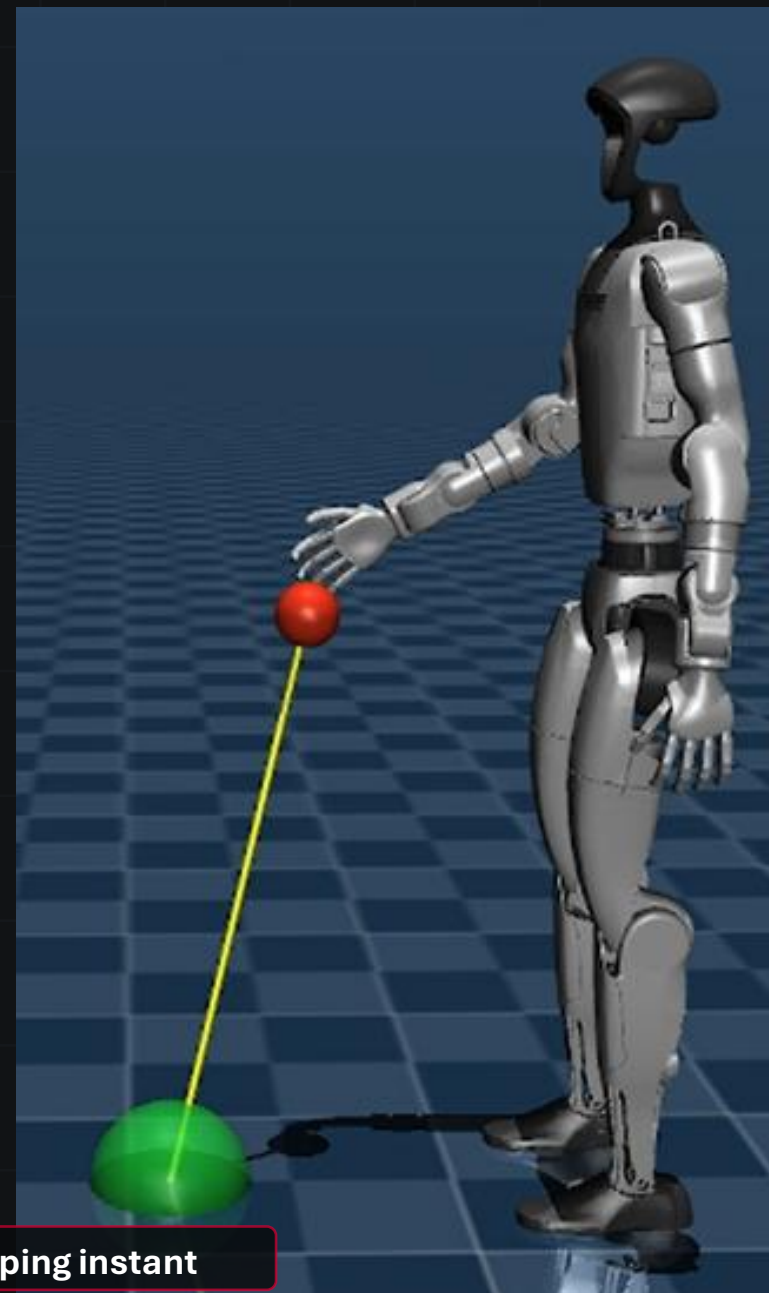
Target

0.55 m ahead

radius 0.1m

MuJoCo

2ms physics timestep



First dropping instant

Three Problems That Nearly Invalidated the Results

A visually successful throw is not automatically a valid experiment.

01

Cause ⚠️

Timestep too coarse

BALL CROSSED THE FLOOR

Fix ✅

Physics micro-stepping

02

Cause ⚠️

Physical joint angles sent into
normalized action space

ARM MOTION CLIPPED

Fix ✅

Joint-limit mapping to $[-1, 1]$

03

Cause ⚠️

Gymnasium termination

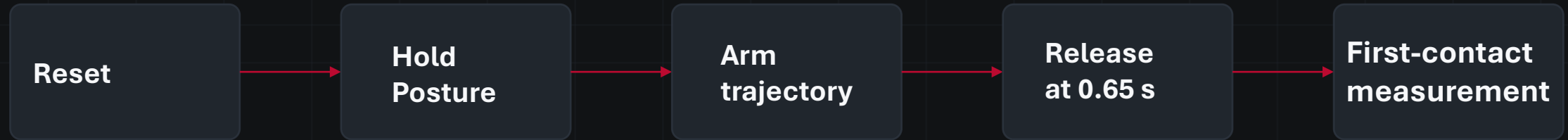
EPISODE ENDED TOO EARLY

Fix ✅

Detached post-contact
diagnostic continuation

The Baseline Was Not the Final Answer

A reference controller—not an optimal controller.



Deterministic engineering benchmark

[Live Running Simulations - LEVEL02](#)

Learning Without Throwing Away What Already Worked

Small corrections around a stable reference.

Scripted baseline
action

+

PPO residual $\times 0.2$
(refining factor)

=

Final joint
command

observation

33-D

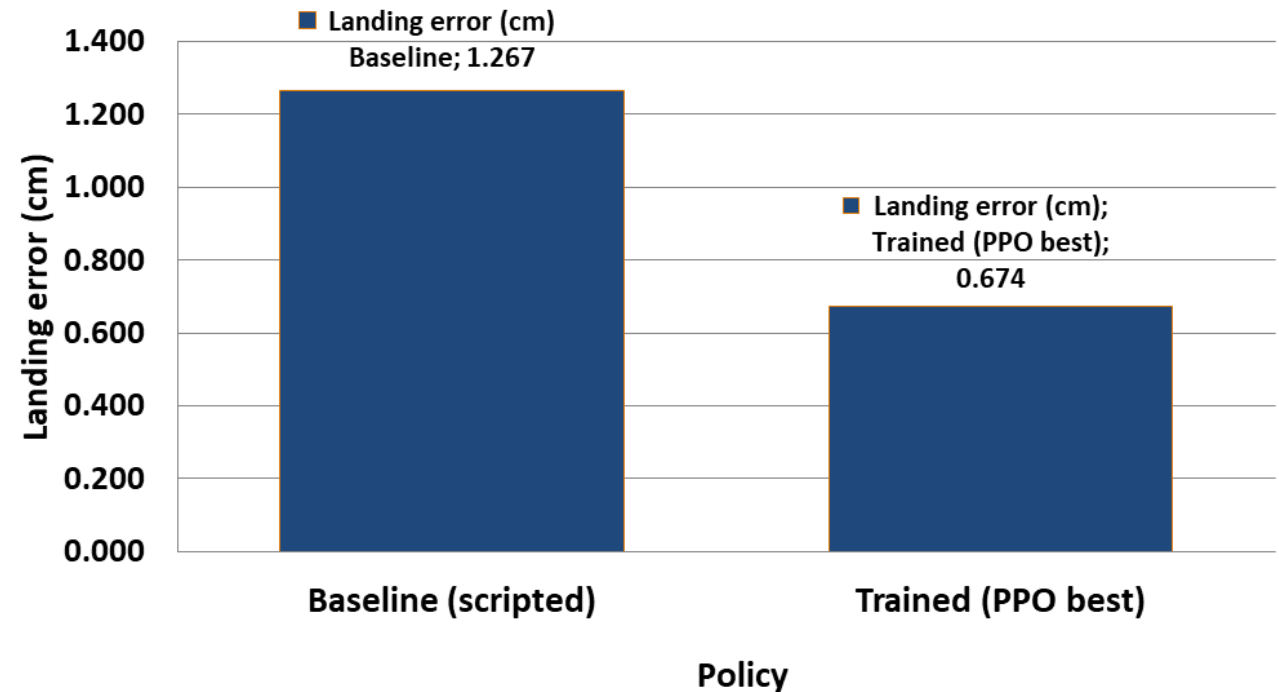
residual action

7-D

[256, 256]

MLP

Mean landing error: baseline vs trained (cm)



How We Prepared for Transfer

Sim2Real analysis was done for both levels; hardware validation remains the next physical proof.

Level 2 = Standing drop

ANALYZED

Transfer risk is mainly contact physics, actuator delay and sensor noise.

Level 3 = Two-steps+jump throw

ANALYZED

Transfer risk also includes locomotion balance and two-hand release timing.

LVL03: Let's Play Basketball

Validated environment contract.

Unitree G1

29 DoF

Ball

radius 4 cm

mass 50 g

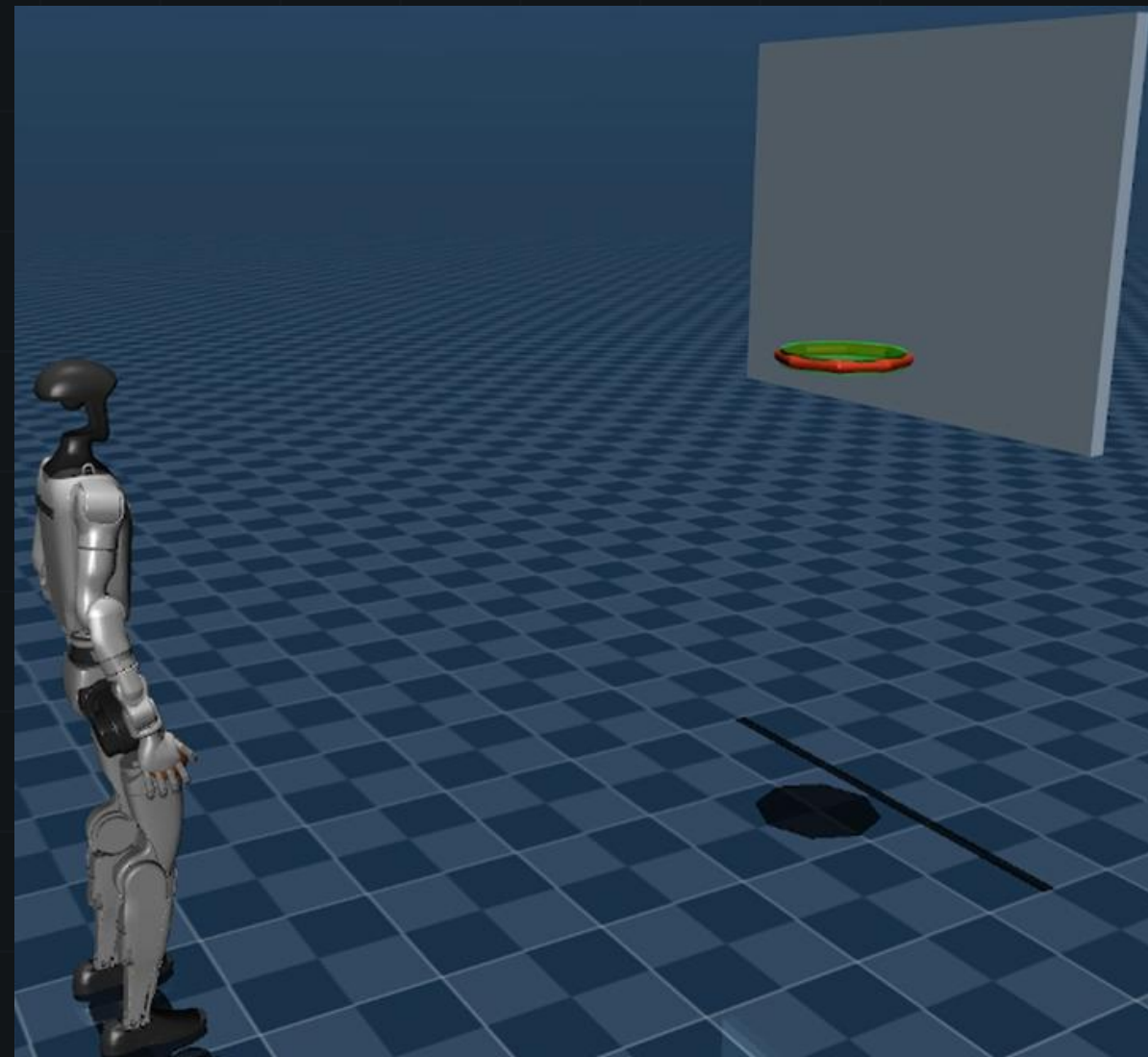
Target

1.8m ahead

1.2m high

MuJoCo

2ms physics timestep



First throwing instant

Three Problems That Nearly Invalidated the Results

A visually successful throw is not automatically a valid experiment.

01

Cause 

No pelvis-joint welded to the ground.

WEIGHT TRANSFER DURING WALKING

Fix 

Implemented a pre-step weight shift aided by a PD Gyroscope.

02

Cause 

Instantaneous joint commands → jerk.

FALLING DOWN DUE TO JUMP

Fix 

Step-functions → linear keyframe interpolation (smoother motion).

03

Cause 

Too restricted action space (8 variables only).

TWO ARMS COORDINATION

Fix 

Enable of the 29 MuJoCo hardware motors.

The Baseline Is Still Not the Final Answer

A reference controller—not an optimal controller.



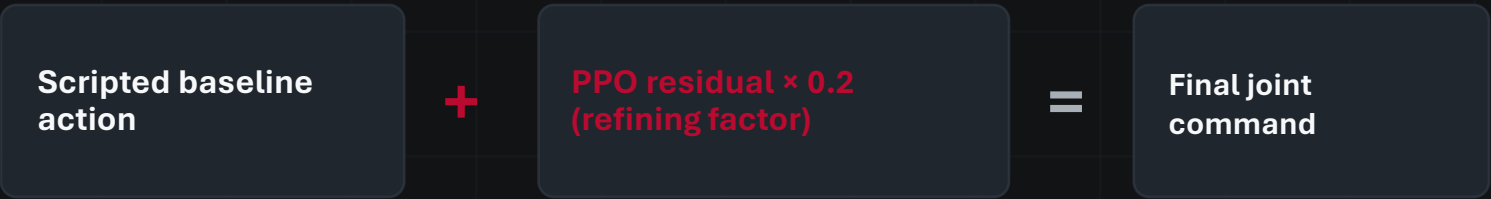
Deterministic engineering benchmark

[Live Running Simulation - LEVEL03](#)

Would You Trust These Results? **We did not.**

Learning Without Throwing Away What Already Worked

Small corrections around a stable reference.



observation

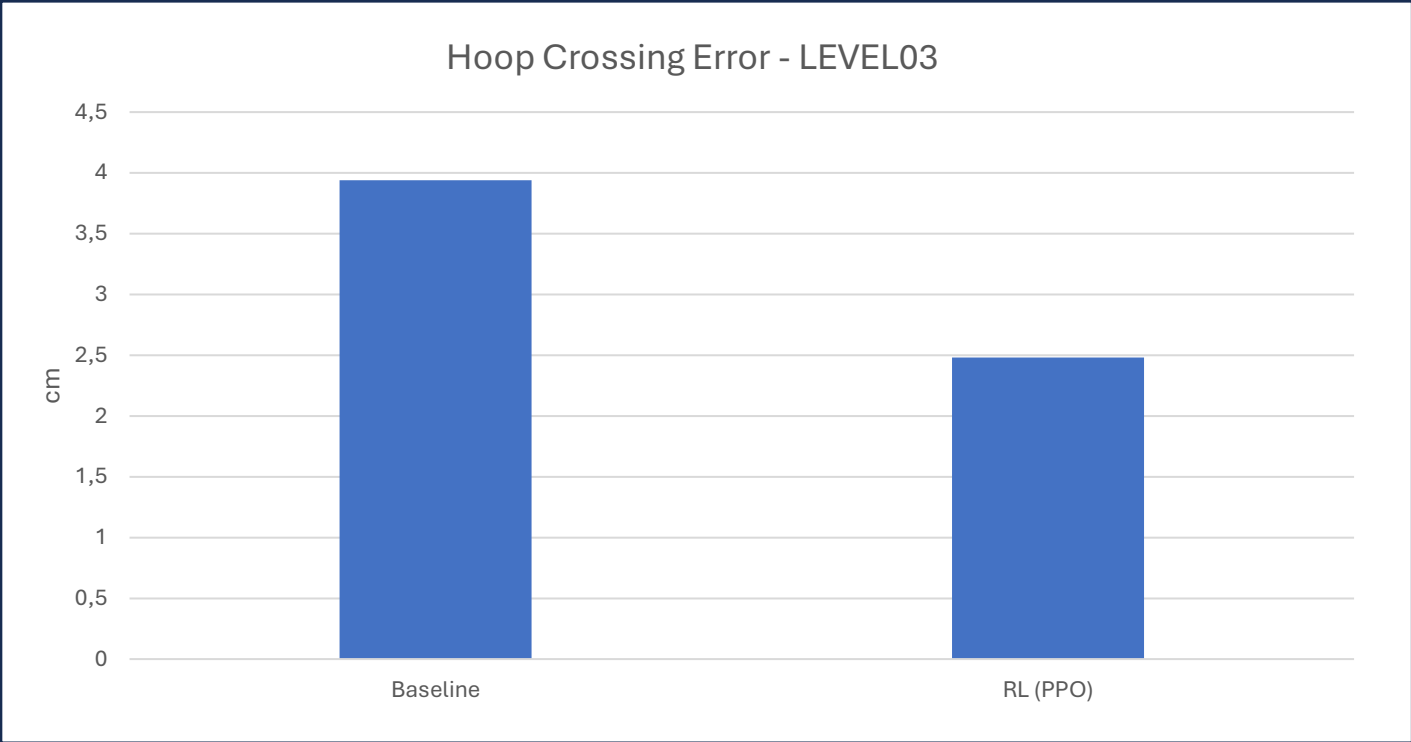
33-D

residual action

7-D

[256, 256]

MLP



Sim2Real

Sim2Real analysis was done for both **levels**; hardware validation remains the next physical proof.

Domain randomization plan

Physics

- **Joint friction:** randomly drawn from [0.7, 1.3] at each reset.
- **Floor friction:** random factor from [0.5, 1.5].

Actuation

Motor strength latency:
3 frames x 2ms = 6ms latency

Sensing

State noise delay:
Physics parameters are randomized at each reset() by directly writing MuJoCo arrays

Task

Self-balanced motion
Proper basketball throw

Transfer workflow

Nominal sim

validated

Randomized sim

robustness

Hardware test

next proof

How Do We Prevent a Biased Comparison in L02 & L03?

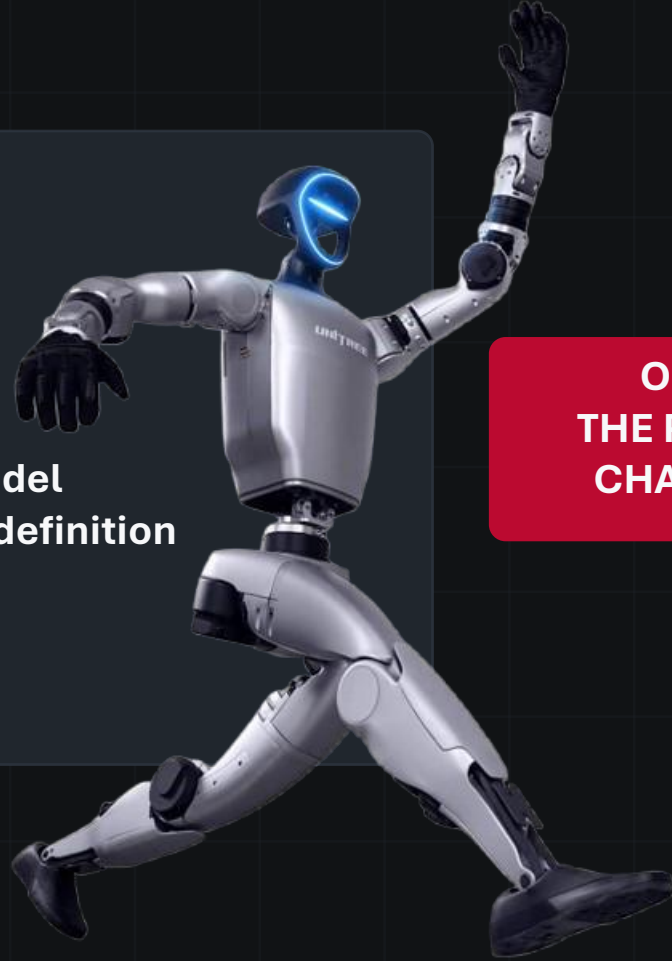
Baseline

- Same seeds
- Same target
- Same world model
- Same success definition
- 100 episodes

**ONLY
THE POLICY
CHANGES**

PPO best (trained)

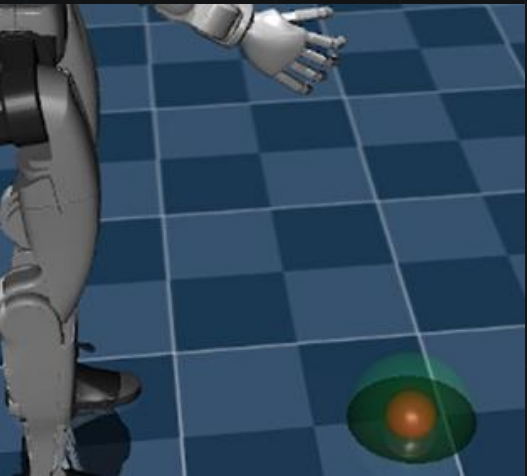
- Same seeds
- Same target
- Same world model
- Same success definition
- 100 episodes



Same Success. Different Precision.

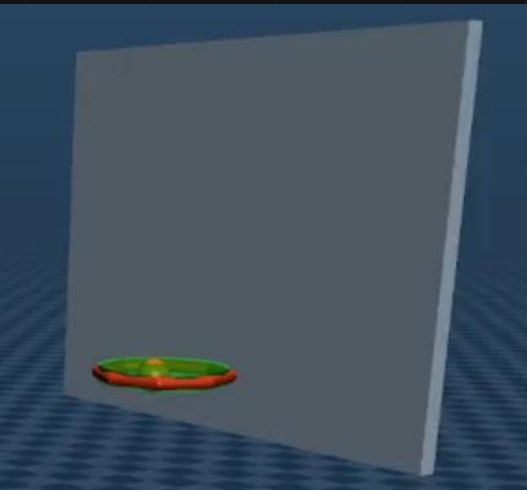
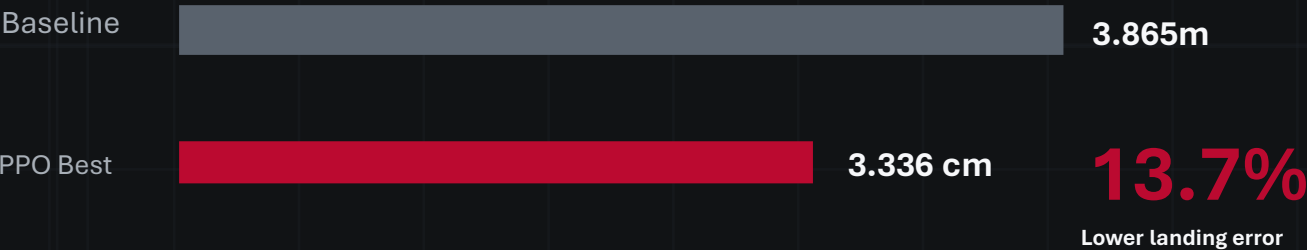
Mean First Contact Error:

L02:



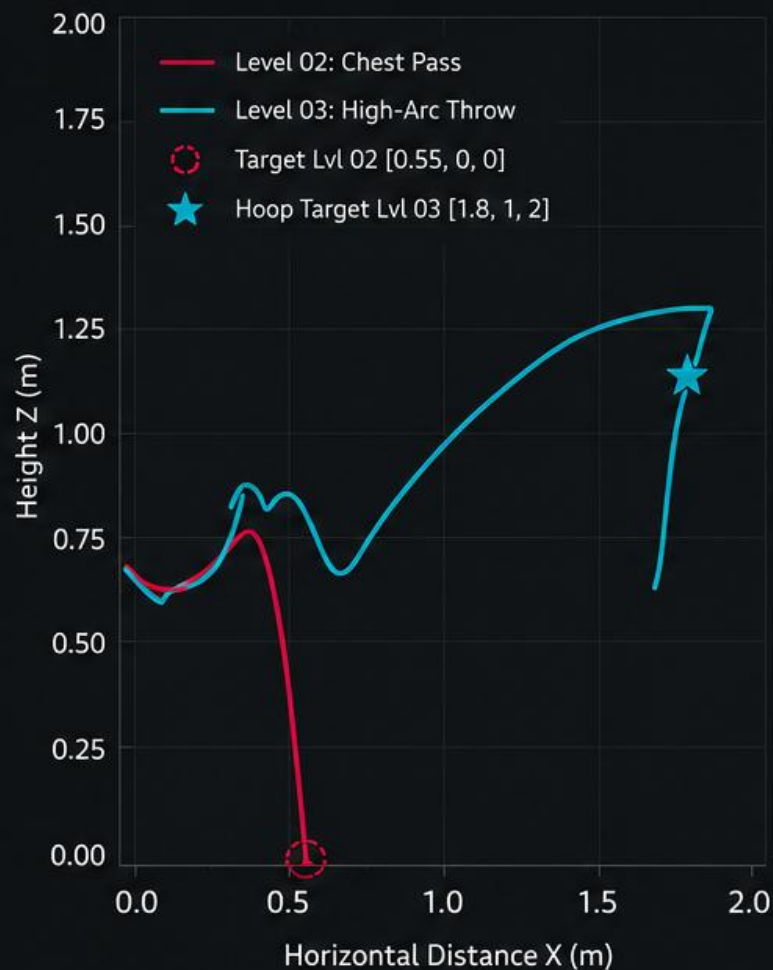
Mean First Contact Error:

L03:

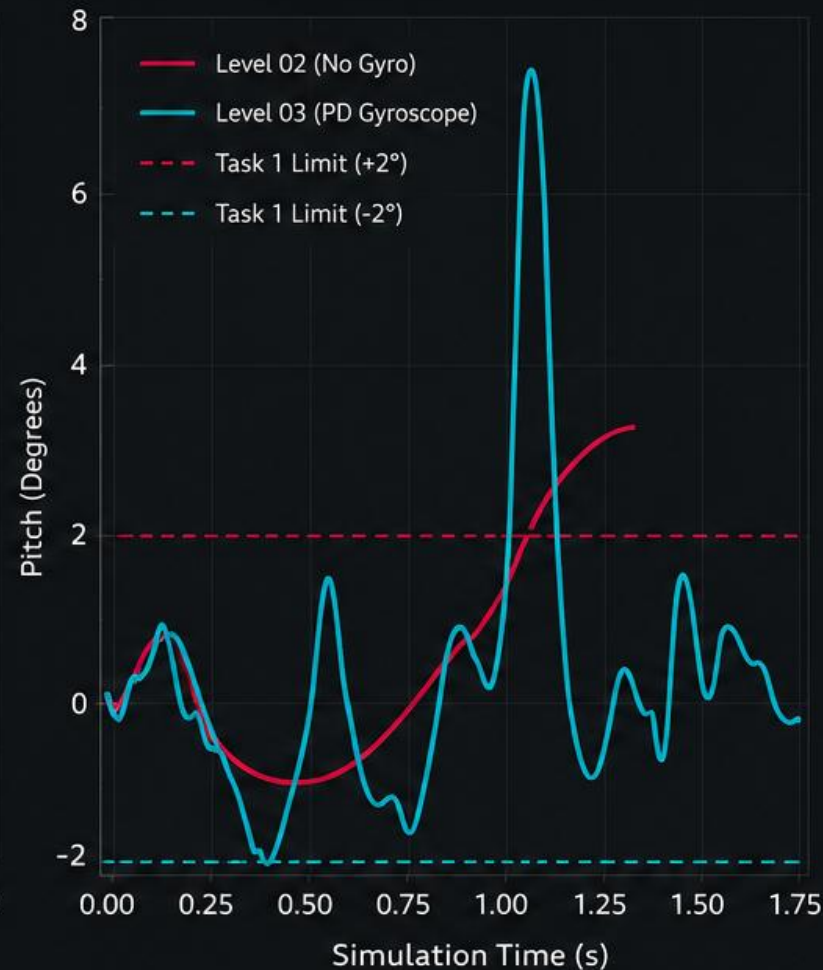


Baseline Evaluation: Level 02 (Drop) vs Level 03 (Pro Throw)

🏀 Ball Flight Trajectory



⚖️ Torso Stability (Pitch Angle)

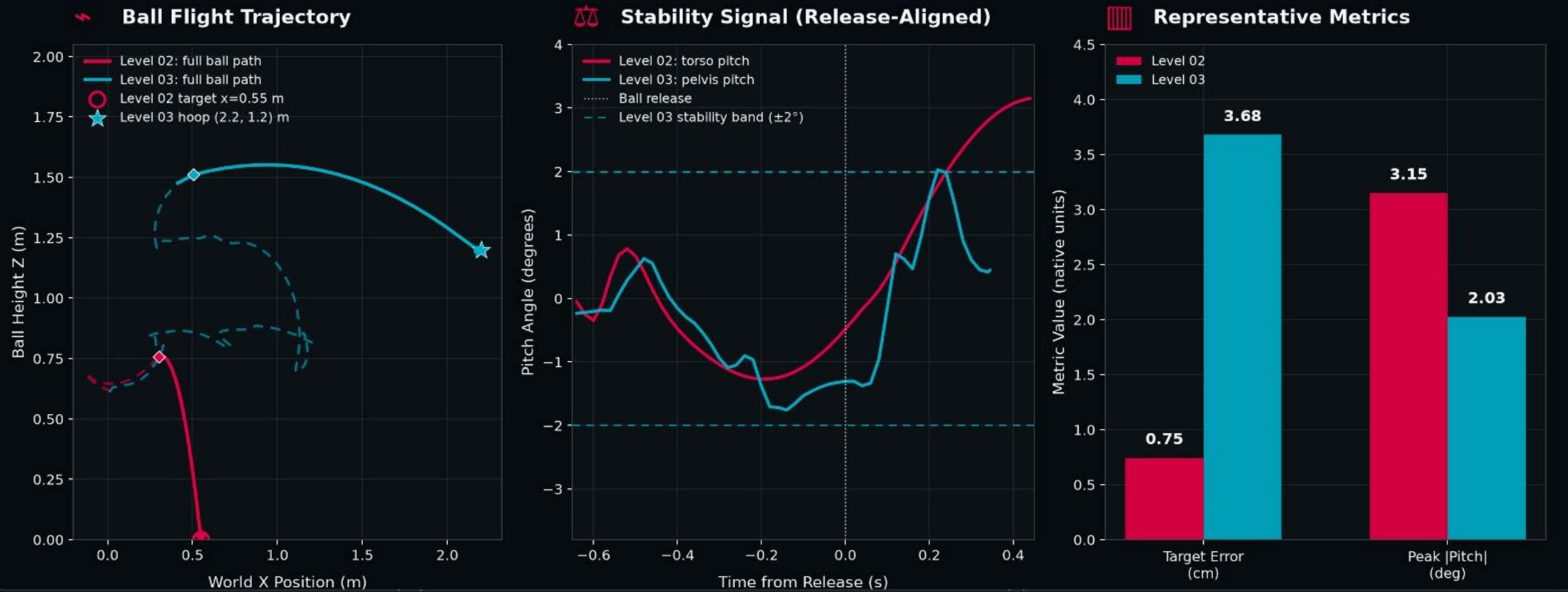


📊 Performance Metrics Comparison



Level 03 (Pro Throw) demonstrates **higher-arc** ball flight, **improved torso stability** within $\pm 2^\circ$ limits, and **lower** final distance **error** compared to Level 02 (Drop).

Policy Evaluation: Level 02 (Fixed-Target Throw) vs Level 03 (PPO Pro Throw)

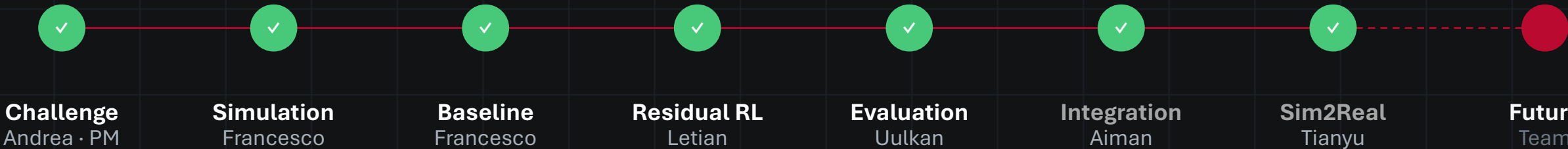


Seed 100000: Level 03 clears the hoop with a 1.55 m apex and 3.68 cm crossing error; Level 02 lands 0.75 cm from its floor target.

The Engineering Journey

● completed ● next

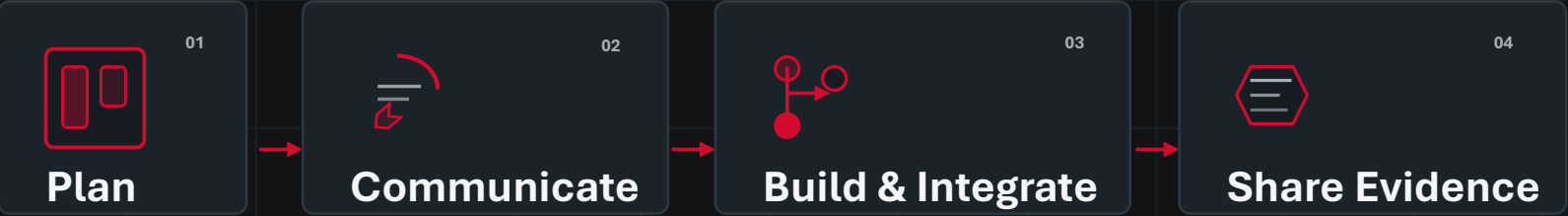
Engineering built reliability. Learning refined precision.



bridge from validated simulation to transfer analysis

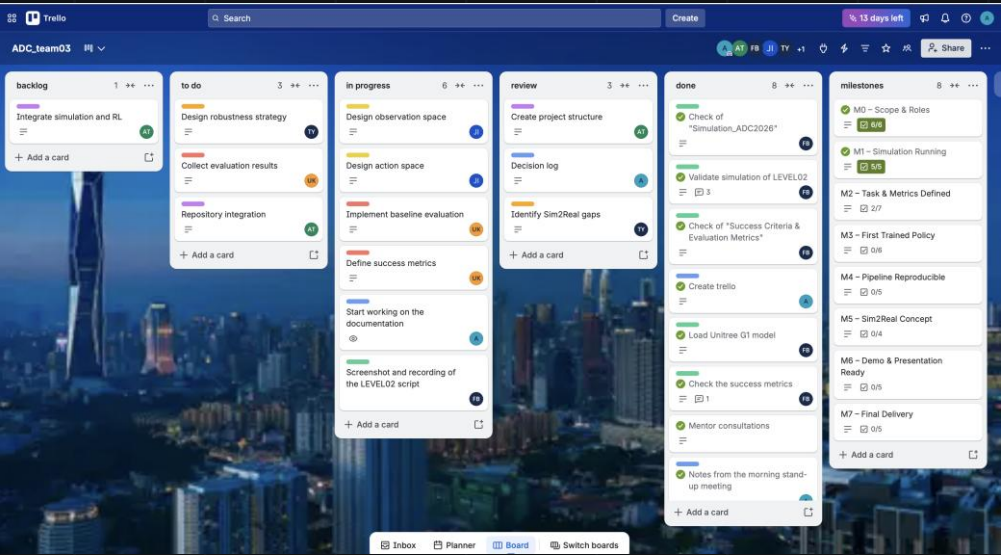
How We Worked Together

From task planning to verified evidence.

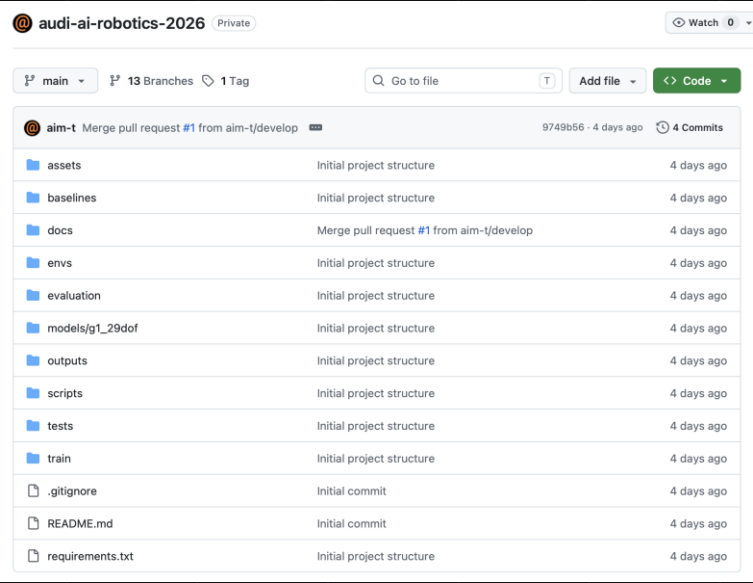
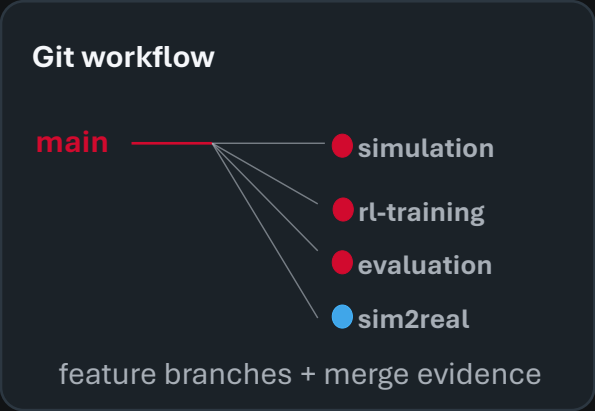


Daily sync
WhatsApp + stand-ups

TRACEABLE EVIDENCE



Trello board — work visible to everyone



Repository — shared code + outputs

One team. One traceable engineering workflow.

Level 2 and Level 3 Final Scope

The same engineering pipeline was executed for both levels.

Level	Simulation	Training	Integration	Evaluation	Sim2Real
Level 2 standing drop	✓	✓	✓	✓	✓
Level 3 2 steps + two-hand throw	✓	✓	✓	✓	✓

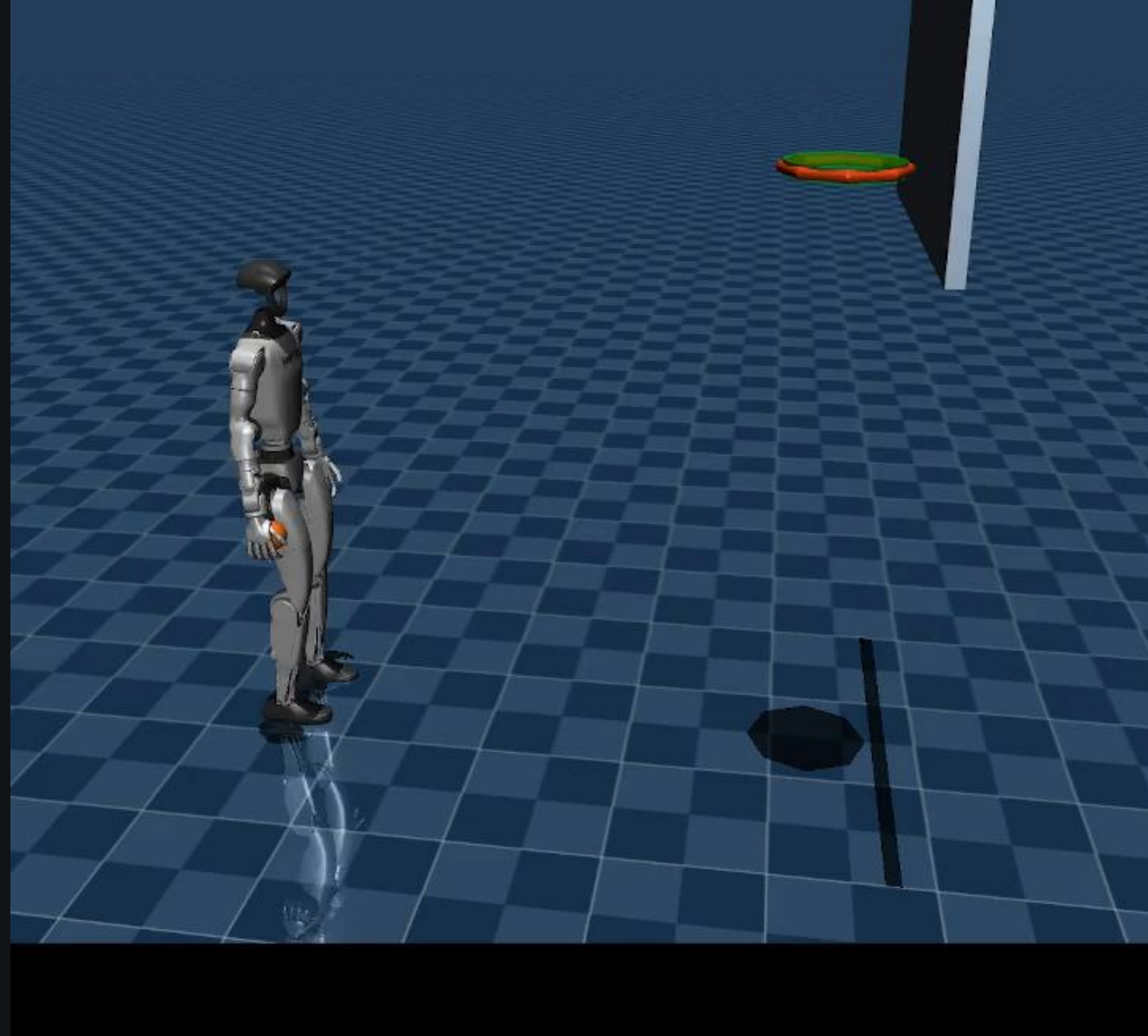
Future Work

From robust simulation to physical validation.



Thank You

Questions?



Trust the simulation. Build the baseline. Teach the robot. Prove the improvement.